

Permutations, Combinations & Binomial Coefficients

Complete worked solutions · All 15 questions · Full method shown

ATTEMPT ALL QUESTIONS

DURATION: 2 HOURS

$P(N,R) = N! / (N-R)!$

$C(N,R) = N! / R!(N-R)!$

BINOMIAL THEOREM

COURSE CSI132 — Discrete Structures II	TOPICS Permutations · Combinations · Binomial	QUESTIONS 15 Questions · Full Solutions
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BEFORE YOU START

Key Formulas Reference

ESSENTIAL FORMULAS		
PERMUTATION (ORDERED) $P(n, r) = n! / (n-r)!$ Order matters. r items from n.	COMBINATION (UNORDERED) $C(n, r) = n! / (r! \cdot (n-r)!)$ Order does NOT matter.	FULL PERMUTATION $P(n, n) = n!$ Arrange all n elements.
BINOMIAL THEOREM $(x+y)^n = \sum C(n,k) x^{n-k} y^k$ k runs from 0 to n.	SUM OF ALL SUBSETS 2^n subsets total Each element: in or out.	ODD-SIZE SUBSETS 2^{n-1} subsets with odd size Half of all subsets are odd-sized.

Basic Permutations & Subsets

1

How many permutations are there of the set {a, b, c, d, e, f, g}?

EASY

- 1 The set has 7 elements. We want to arrange ALL of them → use $P(n, n) = n!$
- 2 $7! = 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 5040$

ANSWER $7! = 5040$ permutations

2

Let $S = \{1, 2, 3, 4, 5\}$. (a) List all 3-permutations. (b) List all 3-combinations.

MEDIUM

(A) 3-PERMUTATIONS – ORDER MATTERS

- 1 $P(5, 3) = 5!/(5-3)! = 120/2 = 60$ ordered triples total
- 2 Sample listing (all distinct orderings count separately):

(1,2,3) (1,2,4) (1,2,5) (1,3,2) (1,3,4) (1,3,5) (1,4,2) (1,4,3) (1,4,5) (1,5,2)
(1,5,3) (1,5,4)
(2,1,3) (2,1,4) (2,1,5) (2,3,1) (2,3,4) (2,3,5) (2,4,1) (2,4,3) (2,4,5) (2,5,1)
(2,5,3) (2,5,4)
(3,1,2) (3,1,4) (3,1,5) (3,2,1) (3,2,4) (3,2,5) (3,4,1) (3,4,2) (3,4,5) (3,5,1)
(3,5,2) (3,5,4)
(4,1,2) (4,1,3) (4,1,5) (4,2,1) (4,2,3) (4,2,5) (4,3,1) (4,3,2) (4,3,5) (4,5,1)
(4,5,2) (4,5,3)
(5,1,2) (5,1,3) (5,1,4) (5,2,1) (5,2,3) (5,2,4) (5,3,1) (5,3,2) (5,3,4) (5,4,1)
(5,4,2) (5,4,3)

TOTAL $P(5, 3) = 60$ permutations

(B) 3-COMBINATIONS – ORDER DOES NOT MATTER

- 1 $C(5, 3) = 5!/(3! \cdot 2!) = 120/(6 \cdot 2) = 10$ subsets

{1,2,3} {1,2,4} {1,2,5} {1,3,4} {1,3,5}
{1,4,5} {2,3,4} {2,3,5} {2,4,5} {3,4,5}

TOTAL $C(5, 3) = 10$ combinations

3

Win, place, show in a horse race with 12 horses. How many possibilities?

EASY

- 1 We choose 3 horses from 12 and **order matters** (1st, 2nd, 3rd are distinct positions)

$$2 \quad P(12, 3) = 12! / (12-3)! = 12! / 9!$$

$$3 \quad = 12 \times 11 \times 10 = 1320$$

ANSWER $P(12,3) = 1320$ possibilities

4

n men and n women in a row, alternating gender. How many arrangements?

MEDIUM

1 For alternating, the pattern is either MWMW... or WMWM.... With equal n , both patterns are possible $\rightarrow$ **2 pattern choices**

2 Arrange n men in n slots: $n!$ ways

3 Arrange n women in n slots: $n!$ ways

4 By multiplication principle: $2 \times n! \times n! = 2 \cdot (n!)^2$

ANSWER $2 \cdot (n!)^2$ arrangements

5

How many subsets with an odd number of elements does a set with 10 elements have?

EASY

1 Total subsets of a 10-element set = $2^{10} = 1024$

2 By symmetry of binomial coefficients: exactly **half** of all subsets have an odd number of elements

3 $C(10,1)+C(10,3)+C(10,5)+C(10,7)+C(10,9) = 10+120+252+120+10 = 512$

4 Equivalently: $2^{10} / 2 = 512$

ANSWER $2^9 = 512$ subsets with odd number of elements

02

QUESTIONS 6 – 8

Coin Flips, Arrangements & Line Problems

6

A coin is flipped 10 times. How many outcomes: (a) total? (b) exactly 2 heads? (c) at most 3 tails? (d) same heads and tails?

MEDIUM

(A) TOTAL OUTCOMES

1 Each flip: 2 outcomes (H or T). 10 independent flips $\rightarrow 2^{10}$

ANSWER $2^{10} = 1024$ total outcomes

(B) EXACTLY 2 HEADS

1 Choose which 2 of the 10 flips are heads: $C(10, 2)$

2 $C(10, 2) = 10!/(2! \cdot 8!) = 45$

ANSWER $C(10, 2) = 45$ outcomes

(C) AT MOST 3 TAILS (I.E. 0, 1, 2, OR 3 TAILS)

1 Sum outcomes with 0, 1, 2, or 3 tails:

2 $C(10, 0) + C(10, 1) + C(10, 2) + C(10, 3)$

3 $= 1 + 10 + 45 + 120 = 176$

ANSWER 176 outcomes

(D) SAME NUMBER OF HEADS AND TAILS (5 EACH)

1 Need exactly 5 heads and 5 tails: $C(10, 5)$

2 $C(10, 5) = 10!/(5! \cdot 5!) = 252$

ANSWER $C(10, 5) = 252$ outcomes

7

8 men and 5 women stand in a line so that no two women are adjacent. How many ways?

HARD

1 **Step 1 — Position the 8 men first.** They can stand in any order: $8! = 40320$ ways

2 **Step 2 — Identify valid slots for women.** With 8 men in a row, there are 9 gaps (before, between, and after the men):

_ M _ M _ M _ M _ M _ M _ M _

← 9 available positions for women

3 **Step 3 — Place 5 women in 5 of those 9 gaps** (order matters since women are distinct): $P(9, 5) = 9!/(9-5)! = 9 \times 8 \times 7 \times 6 \times 5 = 15120$

4 **Step 4 — Multiply** (by multiplication principle): $8! \times P(9, 5) = 40320 \times 15120$

ANSWER $8! \times P(9, 5) = 40320 \times 15120 = 609,638,400$

8

13 softball players. (a) Choose 10 players. (b) Assign 10 positions. (c) At least 1 woman (3 women total).

HARD

(A) CHOOSE 10 PLAYERS FROM 13 (ORDER DOES NOT MATTER)

1 $C(13, 10) = C(13, 3) = 13!/(3! \cdot 10!) = (13 \times 12 \times 11)/(3 \times 2 \times 1) = 286$

ANSWER $C(13, 10) = 286$ ways

(B) ASSIGN 10 DISTINCT POSITIONS FROM 13 PLAYERS (ORDER MATTERS)

- 1 Each position is distinct (pitcher, catcher, etc.) $\rightarrow P(13, 10)$
- 2 $P(13, 10) = 13!/3! = 13 \times 12 \times 11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 = 1,037,836,800$

ANSWER $P(13, 10) = 1,037,836,800$ ways

(C) AT LEAST 1 WOMAN (3 WOMEN, 10 MEN ON TEAM)

- 1 Use complement: Total - (no women at all)
- 2 Ways with NO women (choose 10 from 10 men): $C(10, 10) = 1$
- 3 At least 1 woman = $C(13, 10) - C(10, 10) = 286 - 1 = 285$

ANSWER $286 - 1 = 285$ ways

03

QUESTIONS 9 - 12

Answer Keys, Committees & Strings

9

40 true/false questions, 17 are true. How many different answer keys are possible?

EASY

- 1 An answer key is just choosing which 17 of the 40 questions get "True" (remaining 23 get "False")
- 2 Order of questions can vary, but we choose the SET of 17 that are true: $C(40, 17)$
- 3 $C(40, 17) = 40! / (17! \cdot 23!)$

ANSWER $C(40, 17) = 40! / (17! \cdot 23!) \approx 1.10 \times 10^{10}$

10

7 women and 9 men in a math department. Committee of 5: (a) at least 1 woman; (b) at least 1 woman AND 1 man.

HARD

(A) AT LEAST 1 WOMAN

- 1 Total committees of 5 from 16 people: $C(16, 5) = 4368$
- 2 Committees with NO women (all 5 men): $C(9, 5) = 126$
- 3 At least 1 woman: $4368 - 126 = 4242$

ANSWER $C(16, 5) - C(9, 5) = 4368 - 126 = 4242$ ways

(B) AT LEAST 1 WOMAN AND AT LEAST 1 MAN

- 1 Start from part (a) result: 4242 (has at least 1 woman)
- 2 Subtract committees with NO men (all 5 women): $C(7, 5) = 21$

3 $4242 - 21 = 4221$

ANSWER $4242 - 21 = 4221$ ways

11

Strings of 6 lowercase letters. How many contain: (a) 'a'? (b) 'a' and 'b'? (c) 'ab' consecutive? (d) 'a' left of 'b', all distinct?

HARD

(A) STRINGS CONTAINING THE LETTER 'A' (LETTERS CAN REPEAT)

- 1 Total 6-letter strings: 26^6
- 2 Strings with NO 'a': 25^6
- 3 With at least one 'a': $26^6 - 25^6 = 308,915,776 - 244,140,625 = 64,775,151$

ANSWER $26^6 - 25^6 = 64,775,151$

(B) STRINGS CONTAINING BOTH 'A' AND 'B' (LETTERS CAN REPEAT)

- 1 Inclusion-exclusion: Total – (no 'a') – (no 'b') + (no 'a' AND no 'b')
- 2 $26^6 - 25^6 - 25^6 + 24^6$
- 3 $= 308,915,776 - 244,140,625 - 244,140,625 + 191,102,976 = 11,737,502$

ANSWER $26^6 - 2 \cdot 25^6 + 24^6 = 11,737,502$

(C) 'AB' AS A CONSECUTIVE BLOCK, 'A' BEFORE 'B', ALL LETTERS DISTINCT

- 1 Treat "ab" as a single unit → we now fill 5 positions (the "ab" block + 4 other distinct letters)
- 2 Choose where in the 6 positions the "ab" block starts: 5 possible starting positions (pos 1–5)
- 3 Fill remaining 4 positions with 4 distinct letters from the remaining 24 (not 'a' or 'b'): $P(24, 4) = 24 \times 23 \times 22 \times 21 = 255,024$
- 4 Total: $5 \times 255,024 = 1,275,120$

ANSWER $5 \times P(24, 4) = 1,275,120$

(D) 'A' SOMEWHERE LEFT OF 'B', ALL 6 LETTERS DISTINCT

- 1 Count all 6-letter strings with all distinct letters containing both 'a' and 'b'
- 2 Choose 4 more distinct letters from remaining 24: $C(24, 4)$ ways, then arrange all 6: $6!$ ways → but restrict to 'a' before 'b'
- 3 By symmetry, exactly half of all such strings have 'a' before 'b'
- 4 Total distinct strings with both 'a' and 'b': $C(24, 4) \times 6! = 10626 \times 720 = 7,650,720$
- 5 Half have 'a' before 'b': $7,650,720 / 2 = 3,825,360$

ANSWER $C(24, 4) \times 6! / 2 = 3,825,360$

12

Department: 10 men, 15 women. Committee of 6 with more women than men. How many ways?

HARD

- 1 "More women than men" in a group of 6 means: 4W+2M, 5W+1M, or 6W+0M

- 2 **Case 1 (4W, 2M):** $C(15, 4) \times C(10, 2) = 1365 \times 45 = 61,425$
- 3 **Case 2 (5W, 1M):** $C(15, 5) \times C(10, 1) = 3003 \times 10 = 30,030$
- 4 **Case 3 (6W, 0M):** $C(15, 6) \times C(10, 0) = 5005 \times 1 = 5,005$
- 5 **Total:** $61,425 + 30,030 + 5,005 = 96,460$

ANSWER $61,425 + 30,030 + 5,005 = 96,460$ ways

04

QUESTIONS 13 – 15

Binomial Theorem & Coefficients

Binomial Theorem: $(x + y)^n = \sum_{k=0}^n C(n, k) \cdot x^{n-k} \cdot y^k$. The general term is $C(n, k) \cdot x^{n-k} \cdot y^k$. To find a specific coefficient, set exponents equal to what is required and solve for k .

13

Using the binomial theorem, find the expansion of $(x + y)^5$.

MEDIUM

- 1 Apply the formula with $n = 5$. Compute each $C(5, k)$ for $k = 0$ to 5 :

$$C(5, 0) = 1 \mid C(5, 1) = 5 \mid C(5, 2) = 10 \mid C(5, 3) = 10 \mid C(5, 4) = 5 \mid C(5, 5) = 1$$

- 2 Write each term $C(5, k) \cdot x^{5-k} \cdot y^k$:

$$k=0: C(5, 0) \cdot x^5 \cdot y^0 = 1 \cdot x^5$$

$$k=1: C(5, 1) \cdot x^4 \cdot y^1 = 5x^4y$$

$$k=2: C(5, 2) \cdot x^3 \cdot y^2 = 10x^3y^2$$

$$k=3: C(5, 3) \cdot x^2 \cdot y^3 = 10x^2y^3$$

$$k=4: C(5, 4) \cdot x^1 \cdot y^4 = 5xy^4$$

$$k=5: C(5, 5) \cdot x^0 \cdot y^5 = y^5$$

ANSWER $x^5 + 5x^4y + 10x^3y^2 + 10x^2y^3 + 5xy^4 + y^5$

14

Find the coefficient of x^5y^8 in $(x + y)^{13}$.

MEDIUM

- 1 General term of $(x+y)^{13}$: $C(13, k) \cdot x^{13-k} \cdot y^k$

- 2 We need x^5y^8 , so set exponents: $13-k = 5 \rightarrow k = 8$ (check: y exponent = 8 ✓)
- 3 Coefficient = $C(13,8) = C(13,5) = 13!/(5! \cdot 8!) = (13 \times 12 \times 11 \times 10 \times 9)/(5 \times 4 \times 3 \times 2 \times 1) = 1287$

ANSWER $C(13,8) = 1287$

15

What is the coefficient of $x^{101}y^{99}$ in the expansion of $(2x - 3y)^{200}$?

HARD

- 1 Rewrite: $(2x + (-3y))^{200}$. General term: $C(200,k) \cdot (2x)^{(200-k)} \cdot (-3y)^k$
- 2 Expand: $C(200,k) \cdot 2^{(200-k)} \cdot (-3)^k \cdot x^{(200-k)} \cdot y^k$
- 3 Match exponents to $x^{101}y^{99}$: $200-k = 101 \rightarrow k = 99$
- 4 Substitute $k = 99$:

$$\text{Coefficient} = C(200, 99) \cdot 2^{(200-99)} \cdot (-3)^{99}$$

$$= C(200, 99) \cdot 2^{101} \cdot (-3)^{99}$$

- 5 Since $(-3)^{99}$ is negative (odd power): the coefficient is **negative**

$$\text{ANSWER } C(200,99) \cdot 2^{101} \cdot (-3)^{99} = -C(200,99) \cdot 2^{101} \cdot 3^{99}$$

05

REVISION GOLD

Complete Cheat Sheet

All Formulas — *Quick Reference*

PERMUTATION $P(N,R)$

$$P(n,r) = n! / (n-r)!$$

Ordered selection of r from n

COMBINATION $C(N,R)$

$$C(n,r) = n! / (r! \cdot (n-r)!)$$

Unordered selection of r from n

FULL PERMUTATION

$n!$ arrangements

All n objects in a row

TOTAL SUBSETS

2^n total subsets

2^{n-1} with odd size

BINOMIAL THEOREM

$$(x+y)^n = \sum C(n,k)x^{n-k}y^k$$

$k = 0$ to n

BINOMIAL COEFFICIENT

$$C(n,k) = C(n, n-k)$$

Symmetry property

ALTERNATING ROW

NO TWO ADJACENT

COMPLEMENT COUNTING

$$2 \cdot (n!)^2$$

n men + n women alternating

$$m! \times P(m+1, w)$$

m men first, w women in gaps

$$\text{Total} - (\text{unwanted})$$

Use for "at least 1" problems

Problem-Solving Strategy

- 01 – **Does order matter?** Yes → Permutation $P(n, r)$. No → Combination $C(n, r)$.
- 02 – **"At least 1"** → Use complement: Total – (none of that type).
- 03 – **No two adjacent** → Place others first, count gaps, use $P(\text{gaps, restricted})$.
- 04 – **Binomial coefficient** → Match x and y exponents to find k , then read off $C(n, k)$.
- 05 – **More women than men** in group of n → enumerate each valid split, sum cases.
- 06 – **Strings with a AND b** → Inclusion-exclusion: Total – (no a) – (no b) + (no a and no b).
- 07 – **Symmetry shortcut:** $C(n, r) = C(n, n-r)$. Use whichever is easier to compute.
- 08 – **Double integration:** Integrating f'' gives $f' + c$; integrating that gives $f + cx + d$. Two conditions → solve system.

Question Type → Formula

- Arrange all n elements → $n!$
- Pick r , order matters → $P(n, r) = n!/(n-r)!$
- Pick r , order irrelevant → $C(n, r) = n!/r!(n-r)!$
- At least 1 of type X → Total – $C(\text{without } X)$
- No two X adjacent → $\text{Others!} \times P(\text{gaps, } X)$
- Odd-size subsets of n → 2^{n-1}
- Coeff of $x^a y^b$ in $(x+y)^n$ → $C(n, b)$ where $a+b=n$
- More women than men → Sum valid (W, M) cases